Experiment 4, Part A: Expression and Purification of Candidate Binders

Purpose

Produce purified MEGF8- and MOSMO-targeting binders for functional testing. Selected candidates are cloned into a vector built for secreted production, expressed in Expi293F cells, collected from the culture medium, purified by an affinity method, and assessed for purity and recovery. Size-exclusion chromatography removes aggregates, and concentrations are standardised before any functional testing.

Candidate selection

Candidates are chosen on their fold change in the double-positive population against the matched GB1 control in the Experiment 2 screen. Hits are sequence-verified. Selected hits are re-cloned into a secretion vector for expression; the Experiment 2 screening constructs are not used for expression.

Materials and equipment

- pHL-sec secretion vector (Addgene #99845), which carries a secretion signal sequence and a His tag
- AgeI and KpnI restriction enzymes, T4 DNA ligase and buffer; chemically competent DH5α E. coli; LB with carbenicillin (1:1,000 dilution of stock)
- Expi293F cells and expression medium; PEI-Max (1 µg/µL stock); Opti-MEM
- Shaker for suspension culture and 50 mL culture tubes
- Nickel affinity resin that tolerates culture-medium components (for example Ni Sepharose excel); gravity-flow columns or filter plates
- 1 M Tris pH 8.0; 5 M NaCl; wash buffer (50 mM Tris, 500 mM NaCl, 30 mM imidazole); elution buffer (50 mM Tris, 500 mM NaCl, 300 mM imidazole)
- Size-exclusion chromatography system with an S75 column; PBS; SDS-PAGE materials; spectrophotometer

Procedure: cloning into the secretion vector

1. Obtain the coding sequence of each selected binder, by PCR from the Experiment 2 plasmid or as a synthesised fragment, with ends compatible with the cloning sites of pHL-sec. The insert must not contain internal AgeI or KpnI sites.
2. Digest the vector and the insert with AgeI and KpnI and purify the fragments.
3. Ligate each insert into the vector, in frame with the signal sequence and the His tag.
4. Transform DH5α cells with the ligation, plate on LB with carbenicillin, and grow colonies overnight at 37 °C.
5. Pick colonies, grow them overnight, and prepare plasmid DNA by miniprep (Experiment 2, Part B).
6. Sequence-verify each construct.
7. Prepare transfection-grade DNA for each verified construct by midiprep (Experiment 2, Part B).

Procedure: expression in Expi293F cells

1. Choose the culture volume from the protein needed for Part B, which depends on binder size, the number of concentrations and the number of experiments.
2. Use 1 µg of plasmid DNA and 3 µg of PEI-Max for each mL of culture, with cells at about 2×10^6 viable cells per mL. For a 10 mL culture, use 10 µg of DNA and 30 µL of PEI-Max stock (1 µg/µL).

3. Dilute the DNA and the PEI-Max separately in Opti-MEM (about 3–5% of the culture volume each), combine gently by inversion, and incubate at room temperature for 10–20 minutes.
4. Add the mixture dropwise to the cells while swirling gently, and shake the culture.
5. After about 3 days, spin the culture at 4,000 rpm for 5 minutes and collect the supernatant. Repeat the spin to clear it fully.

Procedure: nickel affinity purification

1. To the clarified supernatant, add Tris pH 8 to 50 mM and NaCl to about 500 mM. Do not add imidazole to the supernatant before binding.
2. Add about 20 µL of nickel resin slurry per mL of supernatant, equilibrated in wash buffer. Mix for at least 1 hour at room temperature, or overnight at 4 °C.
3. Collect the resin in a gravity-flow column or filter plate, and wash with 20 column volumes of wash buffer.
4. Elute with elution buffer in two steps, incubating for 10 minutes each.
5. Concentrate the eluate if needed and filter it before size-exclusion chromatography.

Procedure: size-exclusion chromatography and quality checks

1. Run the filtered eluate on an S75 column on an ÄKTA system in PBS. Collect the main peak, which is the putative monomer, and discard aggregate fractions.
2. Check purity and integrity by SDS-PAGE.
3. Measure protein concentration by absorbance at 280 nm, using the extinction coefficient calculated from the sequence.
4. Adjust all binders to the same concentration in PBS.
5. Aliquot and store at −80 °C.

Expected readout

Yield, purity and recovery for each candidate, and a size-exclusion profile with a main monomer peak. A candidate with a large aggregate fraction or low recovery is flagged before functional testing.

References

- Aricescu AR, Lu W, Jones EY. A time- and cost-efficient system for high-level protein production in mammalian cells. Acta Crystallogr D Biol Crystallogr 62, 1243–1250 (2006). https://doi.org/10.1107/S0907444906029799
- pHL-sec, Addgene plasmid #99845: https://www.addgene.org/99845/